What's new

Some slides from Basel

Terence Tao · 16 February 2020

https://terrytao.wordpress.com/2020/02/16/some-slides-from-basel/

tao2typ · converted to Typst 2026-10-14 (UTC)

I have just returned from Basel, Switzerland, on the occasion of the awarding of the 2019 Ostrowski prize to Assaf Naor. I was invited to give the laudatio for Assaf's work, which I have uploaded here. I also gave a public lecture (intended at the high school student level) at the University of Basel entitled "The Notorious Collatz conjecture"; I have uploaded the slides for that here. (Note that the slides here are somewhat unpolished as I was not initially planning to make them public until I was recently requested to do so. In particular I do not have full attribution for some of the images used in the slides.)

Basel has historically been home to a number of very prominent mathematicians, most notably Jacob Bernoulli, whose headstone I saw at the Basel Minster,



and also Leonhard Euler, for which I could not find a formal memorial, but I did at least see a hotel bearing his name:


HOTEL EULER

Comments (23)

 **Allen Knutson** · 16 February 2020

I don't know about memorials, but Euler's actual grave is in St. Petersburg.

 **Rabe** · 16 February 2020

Did you find Hotel Hilbert as well? Thank you for sharing this!

>  **Anonymous** · 17 February 2020
>
> Is that the one in the song? "Since my baby left me, I've had a new place to dwell; Its floors go to infinity, it's the… Hilbert Hotel".

 **Anonymous** · 16 February 2020

Euler is also known as the solver of the Basel problem
https://en.wikipedia.org/wiki/Basel_problem

 **JosephSugar** · 16 February 2020

The "eadem mutata resurgo" is interesting. Wonder if he simply refers to the spiral or it was his philosophical take on life, nature, the universe, in general. The problem is that mutata is both feminine singular and neuter plural.

 **Gil Kalai** · 16 February 2020

Congratulation to Assaf!

 **Max** · 16 February 2020

Thanks for one more inspiring post! But you should please correct repeated instances of "Radamacher" to (Hans) *Rademacher* (in the Assaf laudatio at least, didn't check whether it appears in the other lecture or in your arxiv.org/1909.03562). (I could imagine the Swiss already made that remark – it really hurts the eyes of a German-speaking reader!)
[Corrected, thanks – T.]

 **Radu Zaharopol** · 17 February 2020

Dear Professor Tao,
Many thanks for your message. Sincerely yours, Radu Zaharopol

 **David Fry** · 17 February 2020

Dear Terry,
What an amazing academic trip regarding Switzerland.Â But the burning question here is, HOW WAS THE CHOCOLATE? Thanks, David
Very Sincerely,
David W. Fry

 **Michel Eve** · 18 February 2020

Dear Prof Tao
I have a naive question regarding the Collatz conjecture. As I am just an amateur, I waited to have a low level presentation of your results to understand the progress made .I have therefore a question on your excellent slides on the subject.
You mention the work of Terras which shows that almost all initial values n iterate eventually to a value less than n and you make the point that, if it had been "all" initial values then the conjecture

would have been proven. This seems to say that the conjecture is proven for all initial values except for a small subset S of initial values for which one cannot say anything.
Then you present you latest result which is tighter but still applies to “almost all” initial values but you claim that this is the closest one can get short of a full proof of the conjecture.
My question is : Do you share the same definition of “almost all” with Terras ? ie: is your subset S the same as Terras’s ? If it is the same then it is a step forward but if it is larger ..
I am sure that I must be wrong somewhere ,I would very much appreciate your comment.

> **Michel Eve** · 18 February 2020
> Sorry I see now that my statement “the conjecture is proven for the subset N-S is wrong as some values less than n may lie in S…

Anonymous · 18 February 2020
Hotel Euler is not named after Leonhard. Abraham Euler established a taverna in 1867 which later became the hotel. But of course maybe Abraham was (far) family of Leonhard, born 1707.
I always enjoy reading about the Collatz Conjecture. Some 25 years ago I tried to solve it as well and created pictures of triangles representing the binary digits of the numbers. Was reminded of this when I read your new “almost all” paper.

Andrew Howe · 20 February 2020
Superb Collatz slides thanks – very accessible for aspiring mathematicians. I will be forwarding to a pre-university friend

James Moriarty · 22 February 2020
So you are using the Samsung Galaxy S9 SM-G960U. Well, I suppose it’s got a reasonably safe operating system – except towards Google Inc.

Islem Ghaffor · 23 February 2020
An Equivalent Problem to the Collatz Conjecture
Click to access Ghaffor.pdf

> **Anonymous** · 28 February 2020
> This is a nice idea to represent the Collatz iteration (originally given in terms of one variable) by using two (or more) variables, since the original (one dimensional) formulation of the iteration as a (deterministic) discrete-time Markov chain may have several equivalent representations using finitely many “state variables” – which may be easier to analyze than the original formulation (which is based on a single “state variable”) – the idea to use the Syracuse iteration (in order to simplify the analysis) is similar, but perhaps there are even better (perhaps multidimensional iterations) equivalent formulations which may have additional “nice properties” (to simplify the analysis) than the Syracuse iteration.
>
> > **Robert Silverman** · 28 February 2020
> > I reply here because I do not know how to start a new thread.
> > I am astonished that there has been no mention of Peter Montgomery’s
> > passing:
> > https://fataldeaths.com/2020/Peter-Montgomery-(mathematician)
> > Peter was a good friend and I am deeply saddened by his death.

 **Michael R** · 28 February 2020

https://iacr.org/people/PeterMontgomery.html

 **Michael R** · 28 February 2020

This is the only obit I found via google search.

 **Martin Lotz** · 2 March 2020

There is a memorial to Euler at the place of his birth, in Riehen (a suburb of Basel): https://images.app.goo.gl/W7x696Z9ieEx4Aj67

This is very close to where my parents live, used to go past this place regularly :)

 . · 3 March 2020

Fed cut 0.5 points. Does the fed know something we don't? Is it trying assimilate 'common knowledge' in your parlance?

collatz standing on shoulders of giants (not!) | Turing Machine · 28 September 2020

[…] the quanta writeup. he also has an excellent "high school" level slideshow refd on his blog, my favorite slide is the squirrel at the end that also seems to ref Creation of Adam by […]

 **john ronand** · 17 November 2020

not sure if we can proof Collatz this way

https://photos.app.goo.gl/w4gGNmocgPBqPziu6